

- Microprocessor
 - VLSI(very large scale integration) circuit.
 - Tiny silicon chip
 - Controlling unit of the microcomputer
 - Contains logical element
 - Use for handling data, performing calculations and carrying out stored instructions, etc.
 - 80% of mp operation are data transfer.
- System
 - Group/collection of a process for one big task
- Computer System
 - Consists of several subsystems interconnected by paths
 - CPU control operation of the computer system by executing a sequence of instructions
 - Instructions stored in computer memory
 - This structure is also referred to as a **stored-program computer**.
 - Computer System is an example of a **Microprocessor-Based System**
- Microprocessor-Based System
 - It is a collection of addressable registers.
- Computer Architecture
 - Attributes of a system visible to a programmer and have direct impact on logical execution of a program
 - Attributes such as, Instruction set, no of bits for data types, I/O mechanism and techniques for addressing memory.
- Computer Organization
 - Operational units and their interconnection that realize the architecture specification
 - Hardware details are transparent to the programmer
 - Such as, control signals, interface b/w the computer and peripheral and memory technology which is used.
- Control Unit
 - It is a clocked sequential machine
 - Control all data transfer (instruction from memory to CPU)
 - Decode instructions
 - Carries out operation specified
 - Transformation of data takes place in ALU
- Analog multiplexer
 - Take any kind of analog signal
 - Selecting a specific signal from many, need a control signal
 - A/D convertor also need control signal because different signal needed different logical calculation
- Features of 8085A mp
 - 8-bit mp
 - N-type Mosfet (NMOS) chip device, consist of approx. 6200 transistors
 - External clock can be up to 12MHz
 - Mp pins consists of a large variety of control signals
 - Extra signals for Interrupt, status and serial I/O which allows much more flexibility
 - **IMP:** The instruction set of 8085A consist of 74 instructions.

- Address Bus of 8085A mp
 - 2^{16} (64kb) memory location can address (AD0-AD7 low pins & A8-A15 high pins)
 - No separate data bus
 - Lower pin data bus are time multiplexed,
 - they carry address,
 - other time they carry data
 - Higher pin purely for address line
 - Until each memory reference is complete, the address information are held on these lines and stored in external latches.
- Input/Output port
 - **Can address 256 input and 256 output ports.**
- Instruction cycle
 - Bar(RESET IN) pin if 0 then program counter is reset to 0
 - When pin returned to 1, content of PC are transfer to Address latch, allows address of first instruction to be executed
 - Program execution in MP starts instruction in memory location 'zero'.
 - 8085A instructions are 1-3 byte in length. 1st byte always contain opcode
 - **Data pointer are general purpose register** that contain address of memory or I/O device for data transfer.
 - Temp register pair WZ is not program addressable
 - Stack support interrupt and subroutine programming.
 - Flag bits are
 - Zero
 - Carry
 - Sign
 - Parity
 - Auxiliary
 - Instruction cycle can have one and more Read/Write operation or reference to memory and I/o device
 - **IMP:** Machine cycle, a byte of data is moved into or out of MP, thus Instruction cycle required number of machine cycles.
 - Number of machine cycles depends on instructions.
 - **IMP:** Total Type of machine cycle in instruction cycle can vary up to 1-5.
 - Mp generates signal on pins to allow distinction b/w machine cycles
 - 3 status signal IO/M, S1(bar(write)), S0(bar(read)) are generated at the beginning of each MC and remain valid for the duration of whole mc
 - MC like Memory Read and Write may occur more than once in instruction cycle
- Machine Cycle
 - Machine cycle is to fetch or move in and out the data from memory or I/o or into memory or i/o.
 - 1 MC is 1byte of data transfer
 - If internal data operation, Read/Write is not needed. For ex, mov b,c.
- T-States
 - T1: A memory or i/o device address is placed on address/data bus
 - ALE is generated

- MC status information is placed on IO/M S0,S1 to define type of machine cycle.
- T2: PC is incremented if machine cycle is part of instruction fetch cycle
 - Ready Line is checked
 - In all MC one of the control strobe RD, WR or INTA (all have bars) make a 1 to 0 transition.
- T3: An instruction byte/ data byte is transfer into MP
 - The active control strobe makes a 0 to 1 transition
- T4: The content of IR (instruction register) is decoded
- T5 and T6: Require in some instruction to complete execution
- Tw (optional State) on adjacent to T2 state also called wait state.
 - This state is entered if ready line (READY with bar) is low.
 - The state of address/data and control signals remain the same
 - Ready pin is again checked whether it has to wait or not until ready pin get high signal
- Memory and I/O synchronization (THE WAIT STATE)
 - Timing of memory and I/O system must match with the microprocessor.
 - Use of high-speed memory, I/O and logic devices but it will increase cost
 - Use low frequency crystal in MP but this will degrade performance of it.
 - Design a logic to introduce wait state.
- Memory Mapped I/O
 - To increase device from 256, memory mapped is used.
 - To use a device having both input and output feature, memory mapped I/O can handle these kinds of devices.
 - I/O devices also have 16 bit which can't go directly to 8bit accumulator
 - Isolated I/O data can't directly go in internal register other than accumulator because I/O is 8-bit, so data first goes to accumulator then using other instruction it goes to register like B.
 - Data from B, C pair also has to go one by one to accumulator then go to Output device.
 - But using memory map this whole process done in single instruction.
 - But disadvantage is memory map take a lot space and decoding, other than these two, there isn't any drawback.
 - All microprocessor is capable of memory mapped i/o.
 - 8085 can use both isolated structures and memory mapped io in a single system.
 - Memory mapped io is not limited to accumulator only.
 - In memory mapped io size of directly addressable memory is decreased
 - Decoding of 16-bit rather than 8-bit port address is required
 - 16-bit io transfer is possible using memory map which reduce program execution time.

○ Isolated I/O	○ Memory mapped I/O
○ Isolated i/o / i/o register are separate from memory	○ Memory mapped i/o treat i/o devices as memory locations

○ Use special input and output instructions (IN, OUT)	○ Can use memory reference instruction as need it transfer data it or from I/o device.
○ No of devices addressable is limited to a few hundred (256)	○ $2^{16} = 64k$ ○ No of devices can be 65536

- Instruction Set

- The 74 instructions are divided into five groups
 - Data transfer group (mov,lda,ldi,sta etc)
 - Logical instruction group (and,or,xor,complement,rotate)
 - Branching instruction group (jmp,call,ret)
 - Stack and machine control instruction group (push,pop,hlt,in,out)
 - Arithmetic instruction group (add,sub,inc,dec)
- **Characteristics of data transfer instruction**
 - These instructions perform data transfer between registers
 - One of the registers is always in the microprocessor.
 - The other register is in one of the following locations
 - Microprocessor
 - Memory
 - I/O devices
 - STA memory address, LDA memory address, IN port address, OUT port address
 - Each data transfer instruction identifies source a destination register.
 - Identification of one on both of these registers may be implied or explicit by the instruction mnemonics.
 - Data transfer involve copying the content of the source register into destination, the content of source remains unchanged.
 - Internal registers involve are frequently implied by instruction but the external register is usually identified by the explicit address that is part of instruction word
- **Branching instruction / transfer of control**
 - JMP, CALL, RETURN and flag registers for conditional control
 - All in notes
- **Logical instruction**
- **Branching instruction**
- **Stack and machine control instruction**
- **Arithmetic instruction**
 - All are in notes

- Subroutine and Macro

- Routine is a collection of instruction (program)
- Subroutine = function (program)(method)
- Subroutine word used in low level language

○ Macro	○ Subroutine
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○ During compilation where MACRO comes then the program will paste the line of its code	○ The code will not be pasted but with "CALL" will transfer the control
○ The length of program will increase at compilation time	○ Will take additional time for "CALL" and "RET" at subroutine ○ Length of code will not increase

- Macro syntax
 - M_LABEL_NAME Macro <list of dummy agreements separated by commas>
 - Macro Reference (calling)
 - M_LABEL_NAME <list of parameters>
 - Macro Expansion
 - For example: Load in register indirect which isn't an instruction so macro is used to perform two instructions inside it
 - LDIND MACRO REG, ADDR
LHLD ADDR
REG, M
MOV
ENDM
 - LDIND B, 1000h
 - LHLD 1000h is total 3 bytes which will have OP, MR, MR goes in wz then MR, MR in HL pair
 - MOV B, M is total 1 byte which will have only OP, MR
- Program Assembly and Testing
 - 3 types of instruction
 - Machine instruction
 - Pseudo instructions (assembler directives)
 - Comment instructions
 - Label (optional): opcode, operand1, operand2 ; comment
 - Delimiters : , ;
 - Assembler directives
 - ORG <expression> (this expression is 16 bits address where program will assemble.
 - ORG is origin, origin point where all the instruction starts assembling.
 - END directive Indicate the assembler that this is the end of the program.
 - Equate or SET
 - name EQV expression value/constant (for making constant)
 - name SET value/constant (for changing values)
 - COUNT SET 10
 - Define Storage (DS)
 - Optional Label: DS size
 - This reserve space of given size first address of that space is assigned to label.
 - Define Byte
 - Places a fixed data value after allocating memory
 - Label : DB list
 - L1: DB 8,9,10,11

- Define word DW optional Label: DW 104
 - Initializes one or more 16-bit integers
- Conditional assembly pseudo directives
 - If expression

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ENDIF

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- Special Purpose Register
 - **Program counter** is 16-bit register.
 - The address where the first instruction is saved in memory should be already in the microprocessor that is program counter.
 - **Instruction Register** (IR) is 8-bit register.
 - Instruction is fetched and goes to IR (address 1000h)
 - Then the instruction is decoded in decoder block and the microprocessor gets to know which kind of operation it has to do now.
 - Program Counter has now the address of where to instruction was got. Then it increments by one. (1001h)
 - **Stack Pointer**, stack memory is used as temporary memory, and top of the address of stack is saved in stack pointer.
 - SP, pointing on some location, if addition of anything in stack result in $sp - 1$.
- All Flags are written like this to get the flag value of 8bit data.

S	Z	X	AC	X	P	X	cy
Sign	Zero		Auxiliary		Parity		carry

- PSW register (Processor Status Word)
 - [Accumulator content] [flag content] saved in PSW as higher and lower bits.